

UNIT 3: Weather Chart Analysis

a. Planetary and Macroscale Circulation Patterns

Introduction to Atmospheric Physics and Dynamics

Before you can analyze weather patterns, you need to understand how the atmosphere flows and circulates. This section introduces the foundational principles that govern air movement, temperature, pressure, moisture, and stability — the building blocks of meteorology. You will explore how different layers of the atmosphere interact with each other to influence weather patterns on a global scale. By mastering these fundamentals, you will gain the tools needed to analyze current weather patterns, anticipate environmental shifts, and eventually predict the future state of the atmosphere.

Cardinal Directions and Relativity

Understanding where weather is and where it is going starts with understanding cardinal directions and relativity. These tools help you accurately describe the location and movement of weather features.

Cardinal Directions

Direction	Symbol	Description
North	N	Points toward the Earth's North Pole; top of most maps
South	S	Opposite of North; points toward the South Pole
East	E	Where the sun rises; right side of most maps
West	W	Where the sun sets; left side of most maps

Compass Degrees

- North = 360°
- East = 90°
- South = 180°
- West = 270°
- Northwest = 315°
- Southeast = 135°

Relativity

Relativity is the ability to look at the relationship and interaction between two values or features without any other outside comparison or fixed standard. Understanding relativity allows you to interpret temperature, pressure, and moisture differences across different regions. For example, if you are trying to compare temperature differences, you say that 40°C is cooler than 70°C but warmer than 30°C . While 40°C might not be considered "hot," relative to 30°C it is warmer, and relative to 70°C it is cooler.

Wind Flow and Direction

Wind is the horizontal and vertical movement of air, driven by pressure differences and influenced by terrain, temperature, and atmospheric structure. Air is evaluated by looking at individual air parcels and how they move horizontally with the wind flow, and vertically within columns of air from the Earth's surface to the top of the atmosphere. As air parcels travel, their size will vary based on the temperature and pressure changes relative to each individual air parcel. Wind terminology provides the language to communicate both the origin and destination of air movement, helping forecasters interpret charts, anticipate upstream impacts, and convey conditions clearly and effectively in meteorological contexts.

Directional Terms

The suffix **-ly** indicates wind origin. The suffix **-ward** indicates wind destination.

Term	Meaning
Southerly	Wind coming from the south, blowing toward the north
Southward	Wind blowing toward the south, regardless of origin
Northerly	Wind coming from the north, blowing toward the south
Northward	Wind blowing toward the north, regardless of origin
Easterly	Wind coming from the east, blowing toward the west
Eastward	Wind blowing toward the east, regardless of origin
Westerly	Wind coming from the west, blowing toward the east
Westward	Wind blowing toward the west, regardless of origin

Wind Speed

Wind speed in the United States is measured in knots (KT). Other units used to measure wind speed in various places around the world are meters per second (m/s), kilometers per hour (km/h), miles per hour (mph). 1 KT is equal to 1.15 mph.

Upstream vs. Downstream

In meteorology, the terms **upstream** and **downstream** are used to describe the relative position and movement of weather features, air parcels, and atmospheric flow. These concepts help forecasters determine where weather is coming from and where it's headed. This information is critical when anticipating impacts, planning operations, and issuing weather warnings. These terms are always relative to your position and the direction of atmospheric flow.

Term	Description
Upstream	The area from which atmospheric flow and weather features are approaching and moving closer.

Downstream	The area from which atmospheric flow and weather features are leaving and moving farther away.
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Atmospheric Pressure Fundamentals

Atmospheric pressure is the force exerted by the weight of the air above a given point. It influences wind, vertical motion, and the behavior of air parcels. Understanding pressure systems and how parcels respond to them is essential for forecasting.

Key Pressure Terms

Term	Description
Mean Sea Level (MSL)	A global reference point (0 ft) used to measure elevation and altitude
Sea Level Pressure	Atmospheric pressure measured at MSL; standard for upper-air analysis
Above Ground Level (AGL)	Altitude measured from the physical surface, which varies with terrain
Station Pressure	Pressure adjusted for the elevation of the reporting station



Units of Measurement

- Inches of mercury (**inHg**): Measured and rounded down to the nearest hundredths (e.g., 29.92 inHg)
- Millibars (**mb**): Measured and rounded down to the nearest tenth (e.g., 1013.2 mb)
- Standard sea level pressure: **29.92 inHg** or **1013.2 mb**

Temperature and Moisture

Temperature and moisture determine the energy, buoyancy, and saturation of air parcels. These variables influence cloud formation, humidity, and precipitation potential — making them critical for weather analysis.

Ambient Temperature

- Measured in degrees
- Meteorologists use Celsius (°C)
- Civilians often use Fahrenheit (°F)

Dew Point and Saturation

- Dew Point: Temperature at which air becomes saturated
- Dew point can never exceed ambient temperature
- Higher dew point = more moisture and higher humidity

Dew Point Depression

- Dew Point Depression: The difference between ambient and dew point temperatures that indicate the level of saturation in an air parcel or a column of air.

Depression Value	Moisture Level
> 5°C	Dry air
1–5°C	Moist air
0°C	Saturated → clouds or precipitation likely

Tropospheric Layers and Vertical Structure

The troposphere is the lowest layer of the atmosphere where nearly all weather occurs. It's divided into zones that influence weather features above and below them. Knowing these layers helps you understand how weather systems evolve vertically.

Layer	Altitude Range	Characteristics
Upper Atmosphere	250mb–700mb (~10,000–35,000 ft)	Contains jet streams, planetary circulation patterns, and features that provide support to the lower levels
Lower Atmosphere	700mb–850mb (~10,000–5,000 ft)	Connects upper-level systems to surface based weather features.
Planetary Boundary Layer	Surface–850mb (~0–5,000 ft)	Influenced by terrain, friction, and surface heating

Parcel Stability and Vertical Motion

Stability determines whether an air parcel will rise, sink, or remain in place. This affects cloud formation and storm development. Temperature will affect an air parcels stability more than anything. Warm air is more unstable than cold air. Unstable warm air rises, while stable colder air will sink. Moisture is another factor in determining parcel stability. Moist air is more unstable than dry air and therefore more likely to rise. Sinking air parcels are considered stable, while rising air parcels are considered unstable. The rate at which parcels rise indicate the amount of instability in a column of air.

Stability Scale

Parcel Type	Stability Level
Warm/Moist	Most Unstable
Warm/Dry	Unstable
Cold/Moist	Stable
Cold/Dry	Most Stable

Atmospheric Heating and Cooling

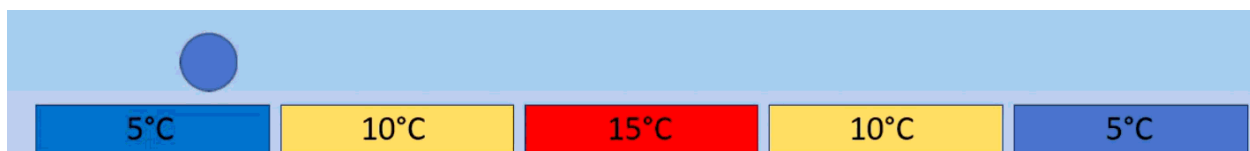
Heating and cooling processes drive daily temperature cycles and vertical motion in the atmosphere. These energy exchanges shape parcel behavior and influence weather intensity, especially in the boundary layer.

Insolation & Eradiation

- **Insolation:** Solar radiation absorbed by Earth during the day
- **Radiation:** Heat released back into the atmosphere at night
- **Maximum temperature:** 1400–1600L (2–4 hours after midday)
- **Minimum temperature:** 0500–0700L (around sunrise)

Conduction & Diabatic Temperature Change

Diabatic temperature change describes the exchange of heat between air parcels due to conduction (direct contact) as they move over relatively warmer or colder surfaces. Warm air parcels will always release heat to become colder, and colder air parcels will always absorb heat to become warmer.



Convection & Adiabatic Temperature Change

Convection describes the vertical temperature change of an air parcel that will either sink toward the surface as it becomes more stable or rise away from the surface as it becomes more unstable. becomes warmer than the surrounding environment and rises away from the surface. The average vertical temperature change, or “lapse rate” is 2°C per 1000 ft.

Adiabatic Cooling: Adiabatic cooling refers to the process by which a rising air parcel releases heat into the atmosphere and decreases in temperature until it reaches equilibrium.

Adiabatic Warming: Adiabatic cooling refers to the process by which a sinking air parcel absorbs heat in the atmosphere and increases in temperature until it reaches equilibrium.

Process	Direction	Heat Exchange	Result
Adiabatic Cooling	Rising	Releases heat	Parcel cools
Adiabatic Warming	Sinking	Absorbs heat	Parcel warms

- Cold air parcels absorb heat, become more unstable, rise away from the surface, and begin adiabatically cooling
- Warm air parcels release heat, become more stable, sink toward the surface, and begin adiabatically warming.

Exercise Questions: Diurnals / Diabatics / Convection

Introduction to Weather Analysis Charts

To display atmospheric conditions, the weather community relies on analysis charts populated with station plots. These plots present wind flow, temperature, dew point, pressure, and height at specific locations for various constant pressure levels. Standard weather charts include the 300mb, 500mb, 700mb, 850mb, Surface, and 1000-500mb thickness. and Surface charts. It's crucial to remember that these pressure levels are not static; they shift seasonally, expanding and rising during the warmer months due to atmospheric heating, and contracting and lowering during the colder months.

Atmospheric Heights

Atmospheric height refers to the vertical distance between the surface and one of the standard pressure charts used to analyze weather. This distance is measured in **decameters (dam)**, where 1 dam = 10 meters.

Height is governed by the average temperature of the air column between the surface and the given pressure value.

- **Warm air expands**, increasing the vertical distance between levels → **higher heights**
- **Cold air contracts**, decreasing the vertical distance → **lower heights**

Surface & Upper Air Station Plots

Wind Data: We observe and analyze the wind speed and direction by using wind barbs that are attached to every station plot. A wind barb has two sections. First, a solid black line will be connecting to a small, enclosed circle which represents the location of the weather station reporting that data. The direction that line is pointing depicts the direction the wind is coming **FROM** based on the cardinal directions previously discussed in this unit. Second, the tail end of the direction line will tell you the wind speed. The speed will be depicted by either solid black lines, or triangles. A triangle represents 50KT, a solid black line represents 10KT, and a black line that is half the length of the 10KT line is 5KT.

EX. A station plot at 500mb with the direction line pointing directly West, with a black triangle, two solid black lines, and a shorter black line would be wind coming from 270° at 75KT.

Upper Air Plots: Station plots provide crucial data for weather analysis. The height of the pressure level, typically displayed with three digits in decameters (tens of meters), is found in the upper right of the station circle. Note that 300 and 500 mb charts are rounded to the nearest decameter, while 700 and 850 mb charts are rounded to the nearest meter. The temperature in degrees Celsius is located in the upper left of the circle; negative temperatures are indicated with a minus sign (-). Finally, the dew point depression, also in degrees Celsius, is positioned in the lower left.

EX. A station plot at 300mb with 924 to the upper right, -12 to the upper left, and 14 to the lower left, would represent a station plot with a height of 924dam, a temperature of -12°C and a dew-point depression of 14 (meaning the dew point temperature would be -26°C).

Surface Plots: The surface station plot model displays key information including wind direction and speed (represented the same way as on upper-air charts), sea-level pressure, temperature, and dew point temperature. Sea-level pressure is located on the upper right of the station circle, represented by three digits to the nearest tenth of a millibar. The temperature, in either Fahrenheit or Celsius, is found on the upper left, with a minus sign indicating negative values. Finally, the dew point temperature is positioned on the lower left.

EX. A station plot at the surface with 013 to the upper right, 75 to the upper left, and 73 to the lower left, would represent a station plot with a pressure of 1001.3mb, a temperature of 75°F and a dewpoint of 73°F.

Exercise Questions: Station Plots

Lines of Equal Value

Using the data gathered from the surface and upper air plots, we can connect certain features that have equal values based on the observations reported over a given location. There are lines on each chart that represent equal values of a specific unit of measurement. The most common lines of equal value are:

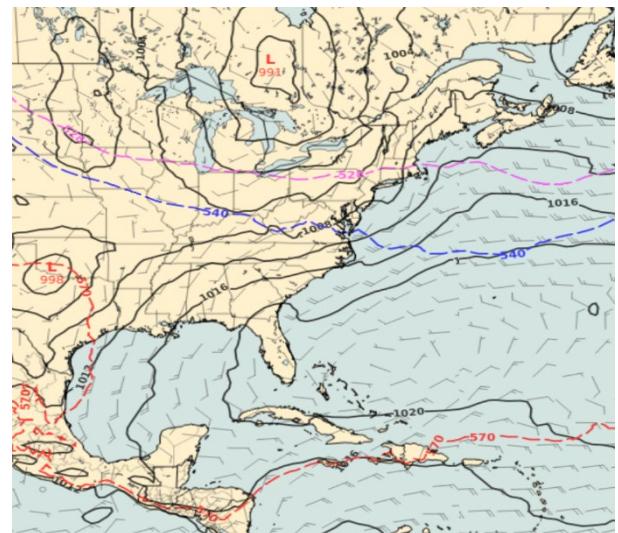
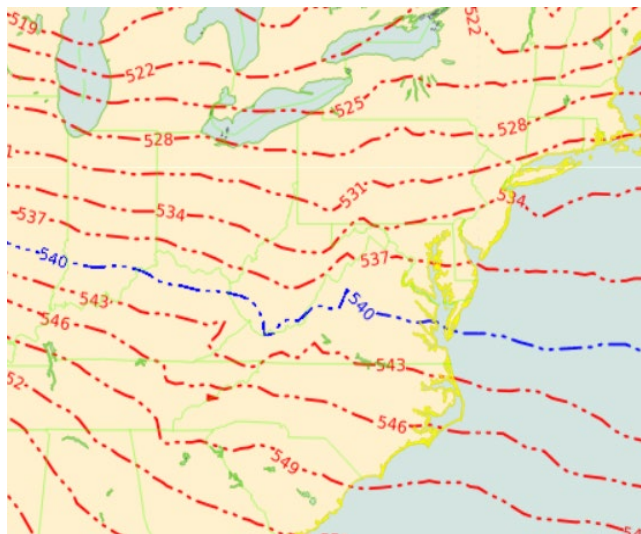
Isotherms: Isotherms are lines of equal temperature. By observing the temperature at different weather stations, we can determine what the line is that separates colder temperatures from warmer temperatures. Isotherms are depicted on charts as dashed red lines.

Isobars: Isobars are lines of equal pressure. Isobars are only found at the surface and are measured in millibars represented by solid black lines. A closed isobar will represent either a high- or low-pressure center which is the basis for all surface-based weather features you will learn later in this unit.

Contours: Contours are very similar to isobars and are also depicted as solid black lines. Where isobars measure lines of equal pressure at the surface, contours measure lines of equal height in either meters or decameters. Just like isobars, a closed contour will represent either a high- or low-height center.

Isotachs: Isotachs are lines of equal wind speed. By taking the wind data from our station plots, we can determine the flow of equal wind speed. Isotachs are represented on a chart by dashed purple lines.

Thickness: Thickness is the vertical distance between two constant pressure surfaces, typically between 1,000 to 500 mb. It differs from contours because it includes a function of temperature and moisture to compute the distance. Thickness lines are depicted as dashed red lines (540 may be in blue). High thickness values indicate warm air; low thickness values indicate cold air.



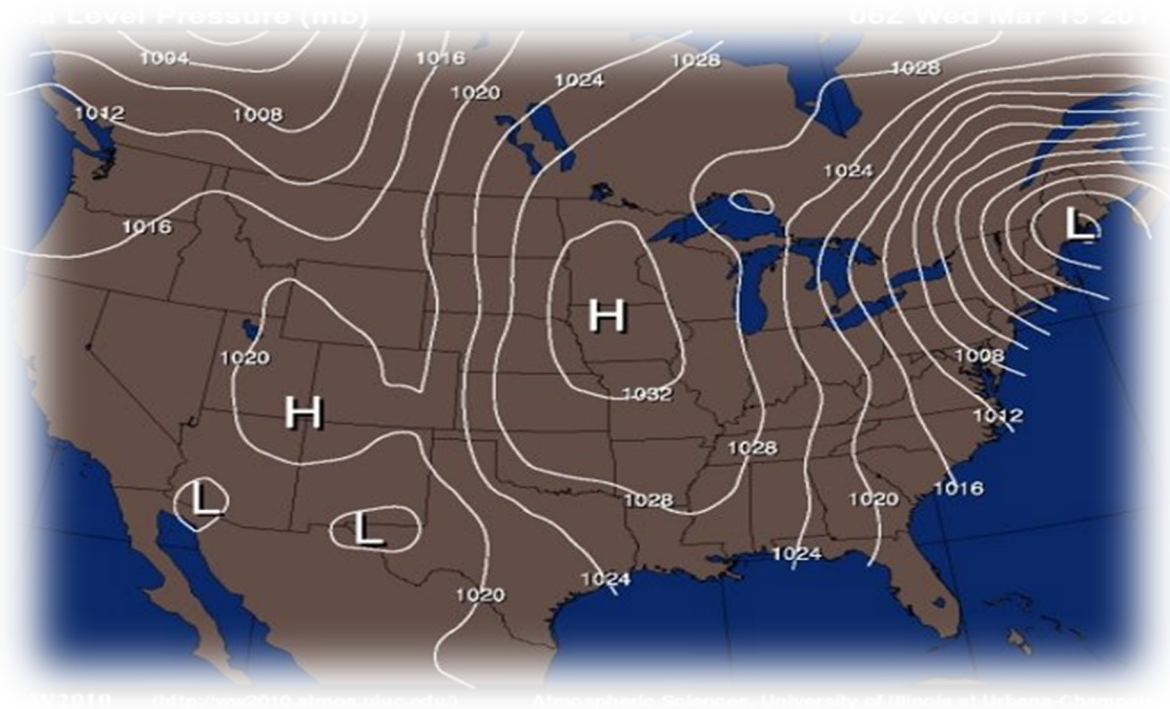
Forces Affecting Parcel Movement

The atmosphere is always in motion due to differing atmospheric physics and forces. Air parcels move horizontally and vertically in a fluid-like motion in a straight-line or curved wind pattern.

Gradient Force

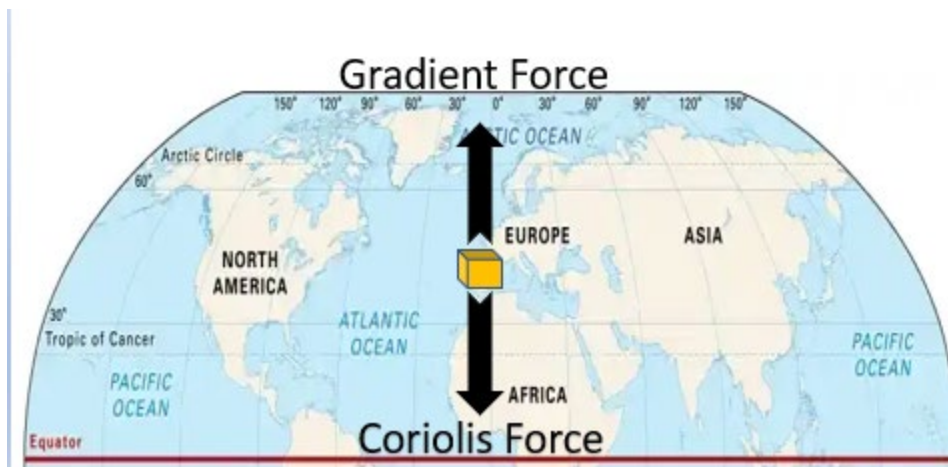
A gradient describes the rate of change in value over distance. A weak gradient has a smaller change in value over a certain distance, whereas a strong gradient has a greater change in value

over the same distance. Gradient force determines wind speed in the upper and lower atmosphere. When gradient force increases, wind speeds also increase. When gradient force weakens, wind speeds will decrease. There are two types of gradient force: Contour Gradient and Pressure Gradient. Contour gradient force initiates winds at any level above the surface and pressure gradient force initiates winds at the surface. Gradient Force is stronger around a low height/pressure center and weaker with a high height/pressure center.



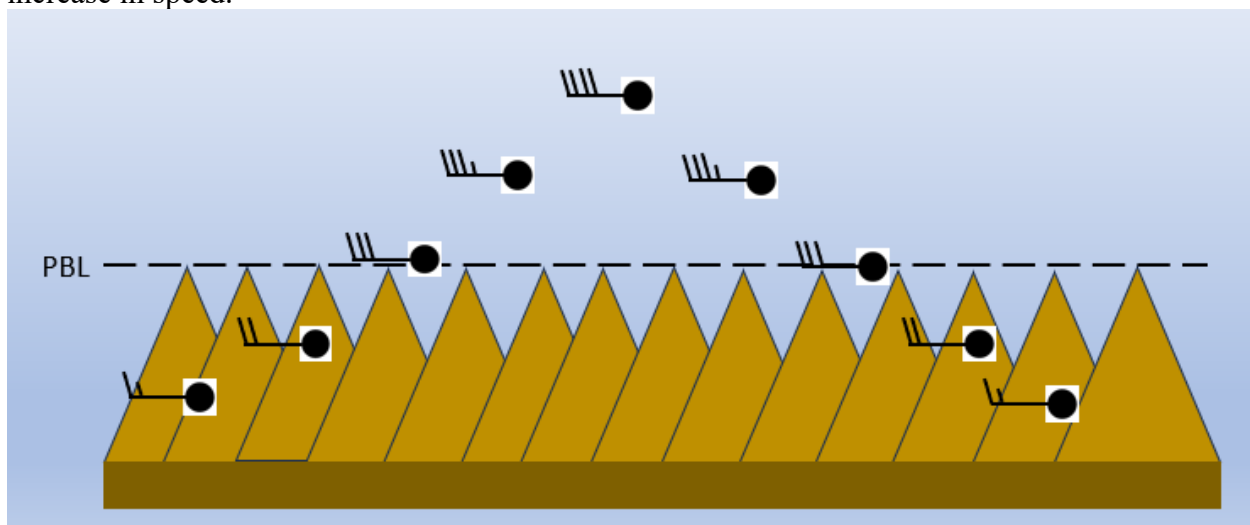
Coriolis Force (CoF)

The opposing force of gradient force is Coriolis Force. Coriolis Force is an apparent force that deflects moving objects (like air flow) due to the Earth's rotation. In the northern hemisphere, this deflection is to the right of the relative wind flow. Its strength varies with latitude; it's strongest near the poles and weakest at the equator. The PGF and Coriolis force typically exist in a state of balance, so as one increases or decreases, the other adjusts inversely to maintain that equilibrium.



Frictional Force

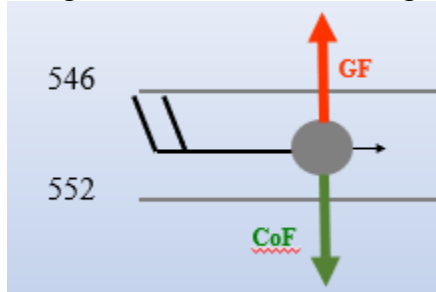
Frictional force opposes parcel movement, and it only exists between the Earth's surface and the boundary layer. Increasing terrain increases friction and decreasing terrain decreases friction. An increase in friction results in a decrease in speed where a decrease in friction results in an increase in speed.



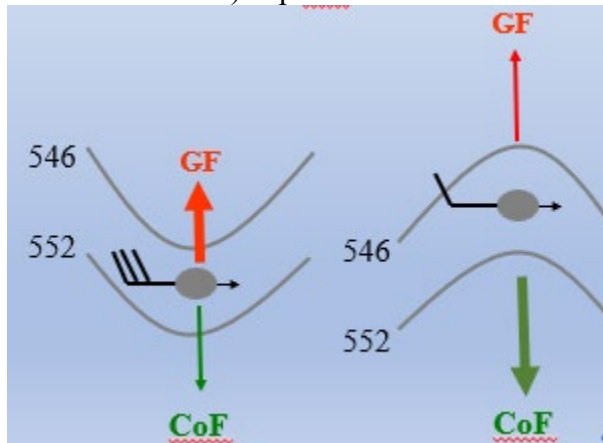
Analyzing Atmospheric Flow

Straight Line vs. Curved Flow

When the Gradient and Coriolis force are equal and opposite, the air is in balance. This results in straight-line flow without changing latitude.



When the Gradient force and Coriolis force are not equal, the atmosphere is in curved flow. This imbalance causes air parcels to rotate and deflect. The direction of deflection (clockwise or counterclockwise) depends on the relative strengths of the forces.

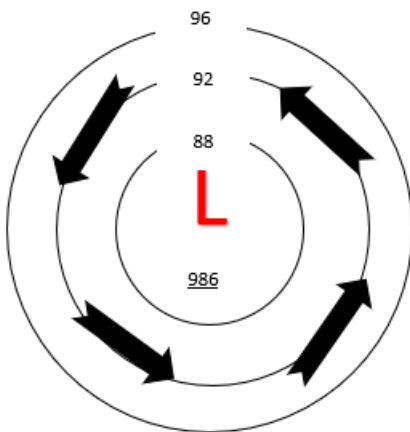


Cyclonic Circulation

Definition: Counterclockwise rotation of air parcels. Occurs when the Gradient force is stronger than the Coriolis force, causing air to deflect to its left.

Trouthing: Characterized by open cyclonic contours in the upper atmosphere, visually represented as a solid black line indicating the axis of the trough.

Surface Low Pressure Center: Defined by closed isobars at the surface, with lower pressure values at the center surrounded by higher pressure. Depicted on weather maps with a red "L".



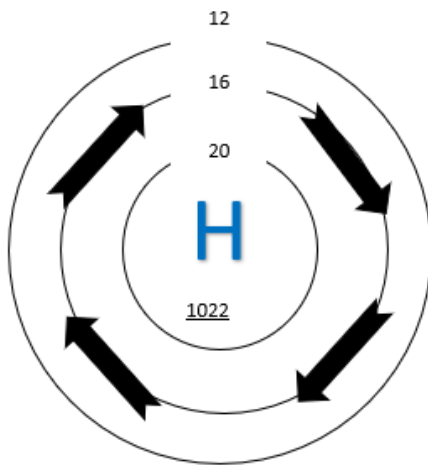
Low Height Center: Defined by closed height contours aloft, with lower height values at the center surrounded by higher heights. Depicted on weather maps with a red "L".

Anticyclonic Circulation

Definition: Clockwise rotation of air parcels. Occurs when the Gradient force is weaker than the Coriolis force, causing air to deflect to its right.

Ridging: Characterized by open anticyclonic contours in the upper atmosphere, visually represented as a jagged black line indicating the axis of the ridge.

Surface High Pressure Center: Defined by closed isobars at the surface, with higher pressure values at the center surrounded by lower pressure. Depicted on weather maps with a blue "H".



High Height Center: Defined by closed height contours aloft, with higher height values at the center surrounded by lower heights. Depicted on weather maps with a blue "H".

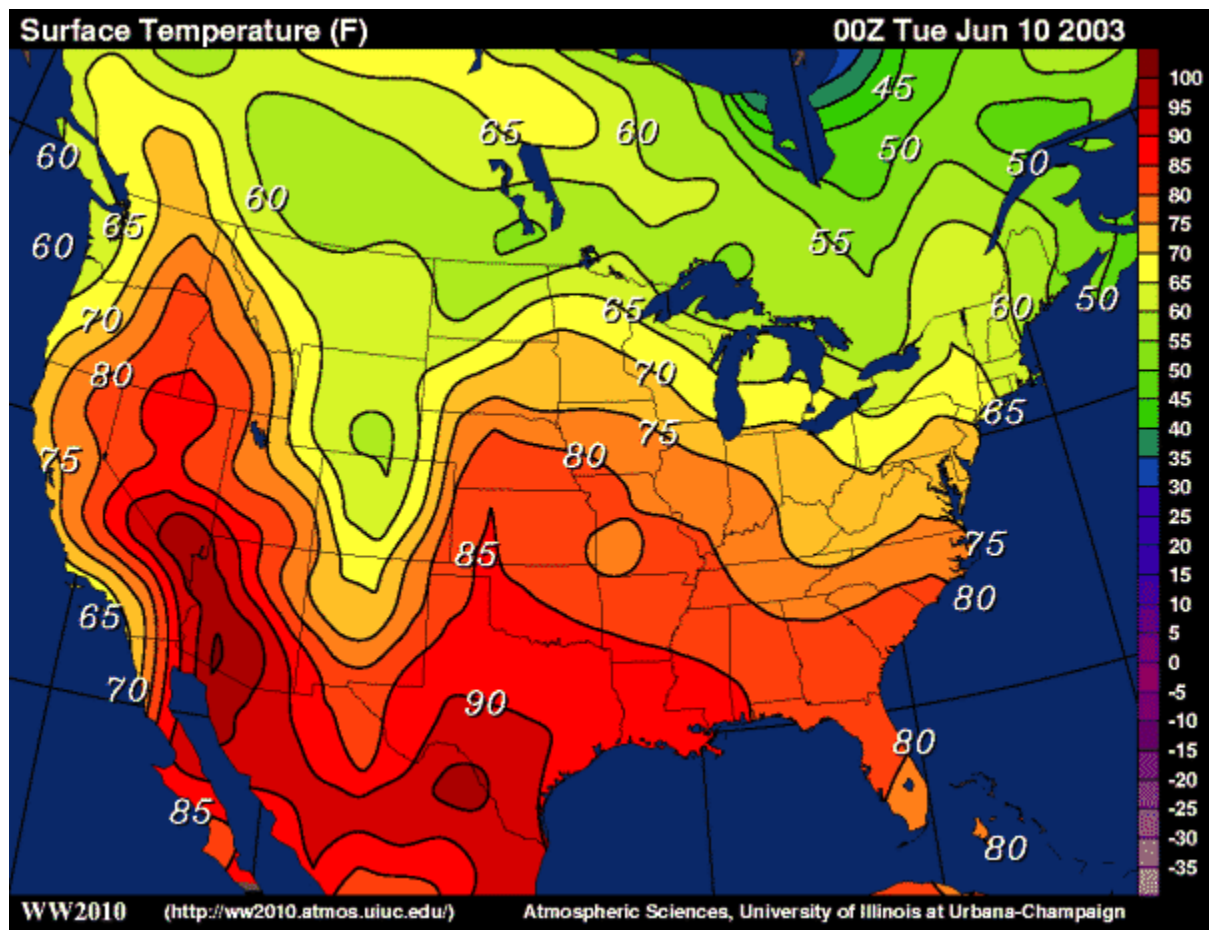
Exercise Questions: Lines of Equal Value / Atmospheric Forces / Atmospheric Flow

Dynamic Processes

Advection

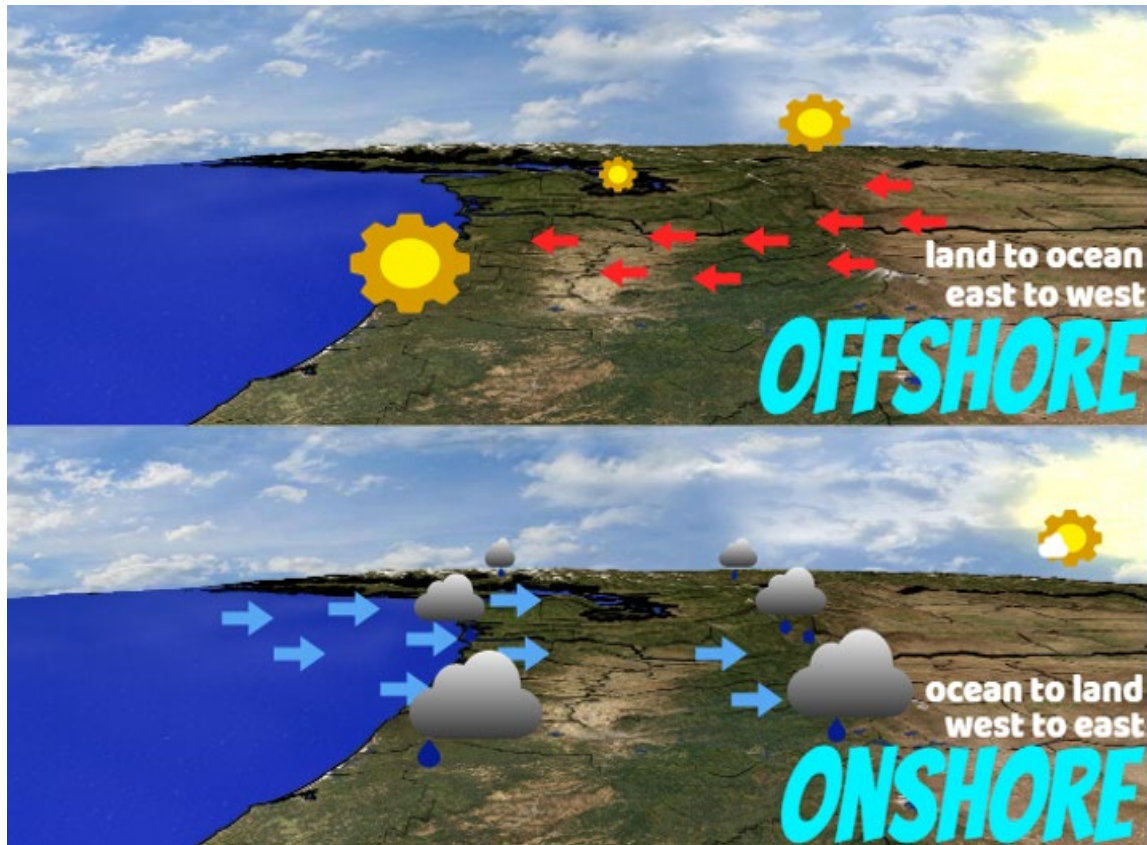
Advection is a change in temperature or moisture due to horizontal movement of air over a specific location.

Temperature Advection: Temperature advection is one of the most important causes of stability change in the atmosphere. It is categorized as either warm air or cold air advection. Warm air advection (WAA) signifies warmer temperatures moving into an area of relatively colder air such as warm air from the equator moving toward the poles. The opposite exists with cold air advection (CAA) which is colder air moving into an area of relatively warmer air such as when cold air from the poles move toward the Equator. Warm air advection indicates decreasing stability in the lower atmosphere, where cold air advection indicates increasing stability. Temperature advection is most seen on analysis charts by a thermal gradient measured by the distance between isotherms. Thermal troughing indicates colder air flowing toward the equator, and thermal ridging indicates warm air flowing toward the poles.



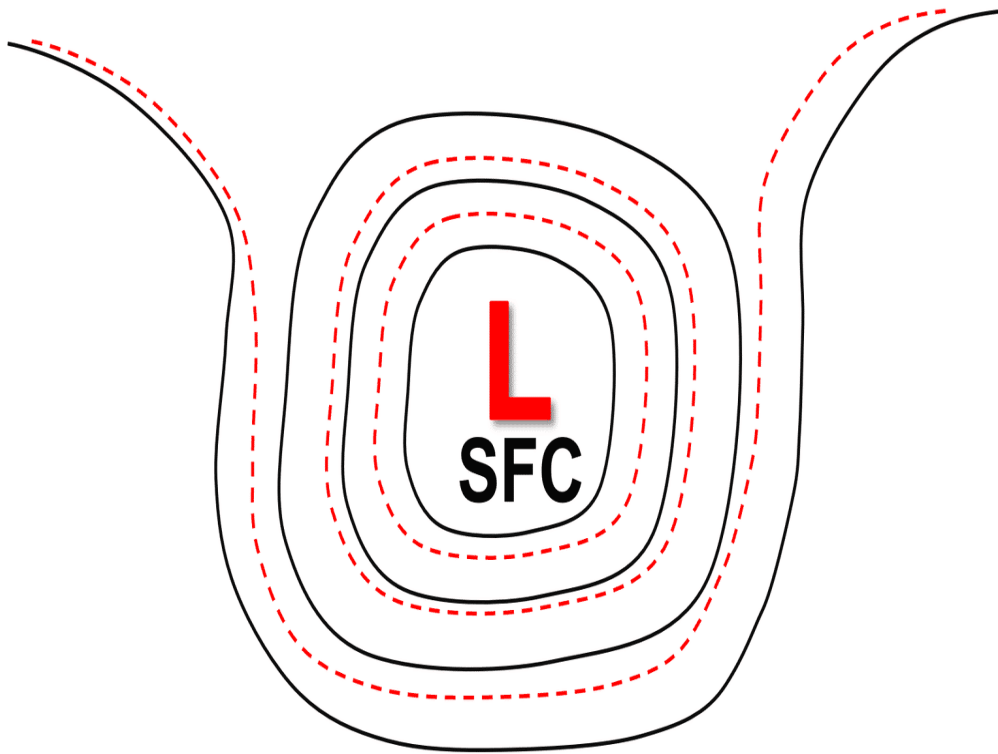
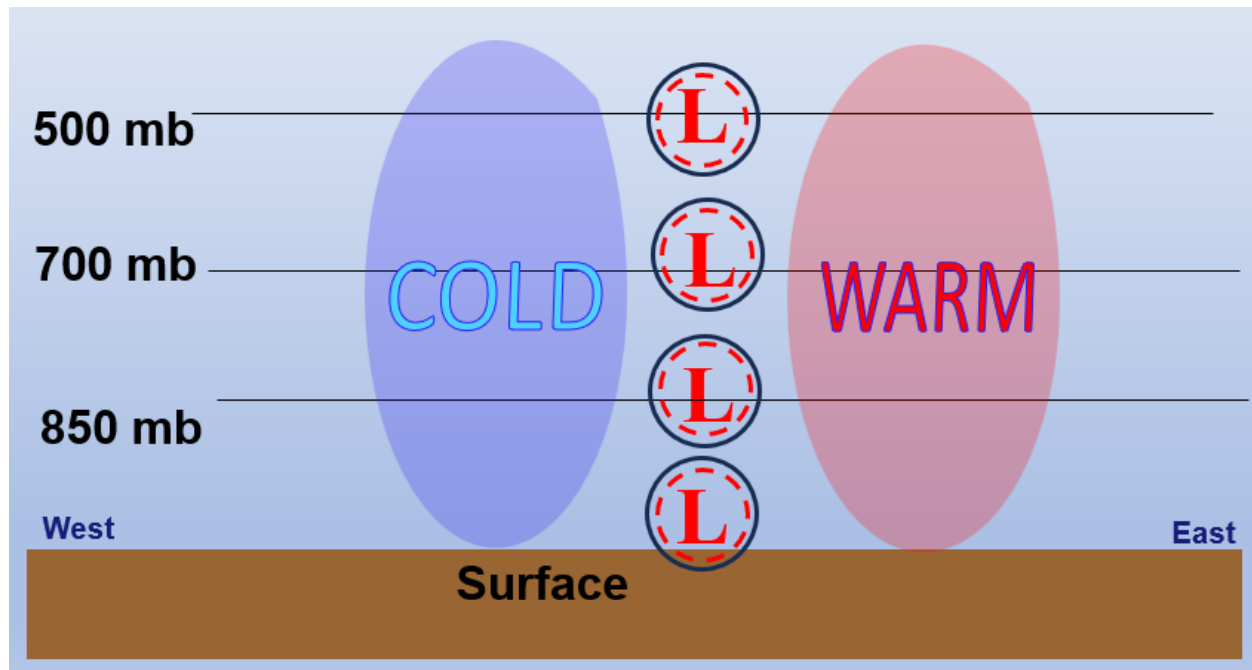
Moisture Advection

Changes in dew-point temperature and dew-point depression signify dry or moist air advection. Dry air advection occurs when dry, continental air moves into an area of moisture like the dry air over land moving over a body of water. Moist air advection occurs when moist, maritime air moves over a dry surface like air from a body of water moving onshore and over the land.



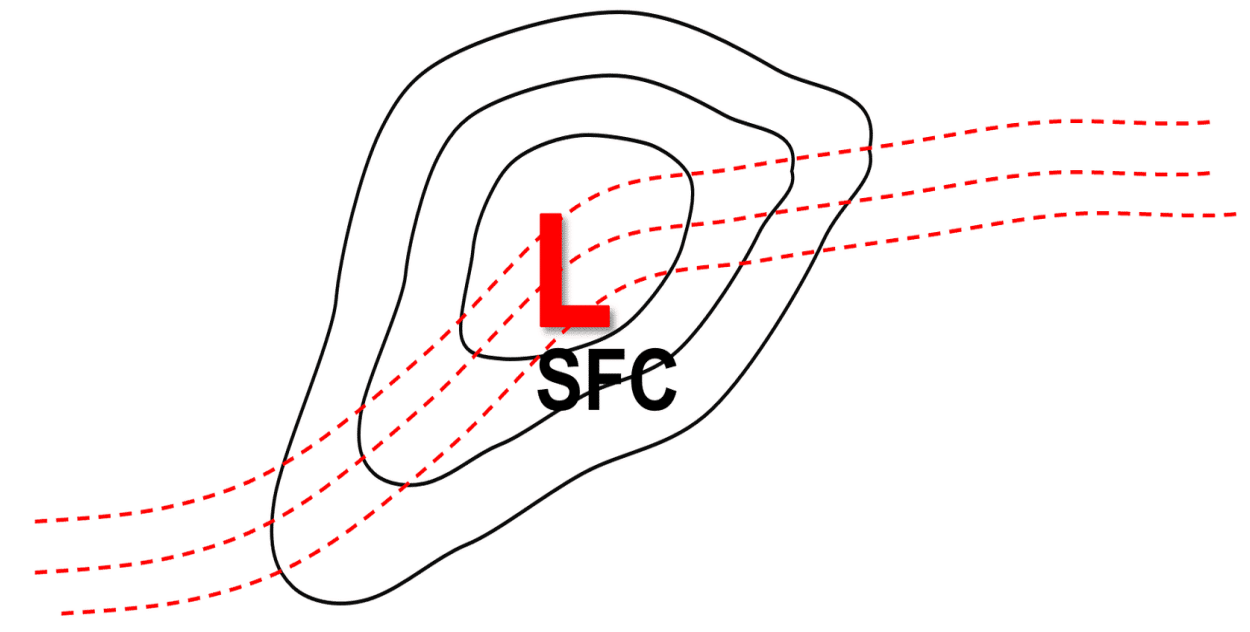
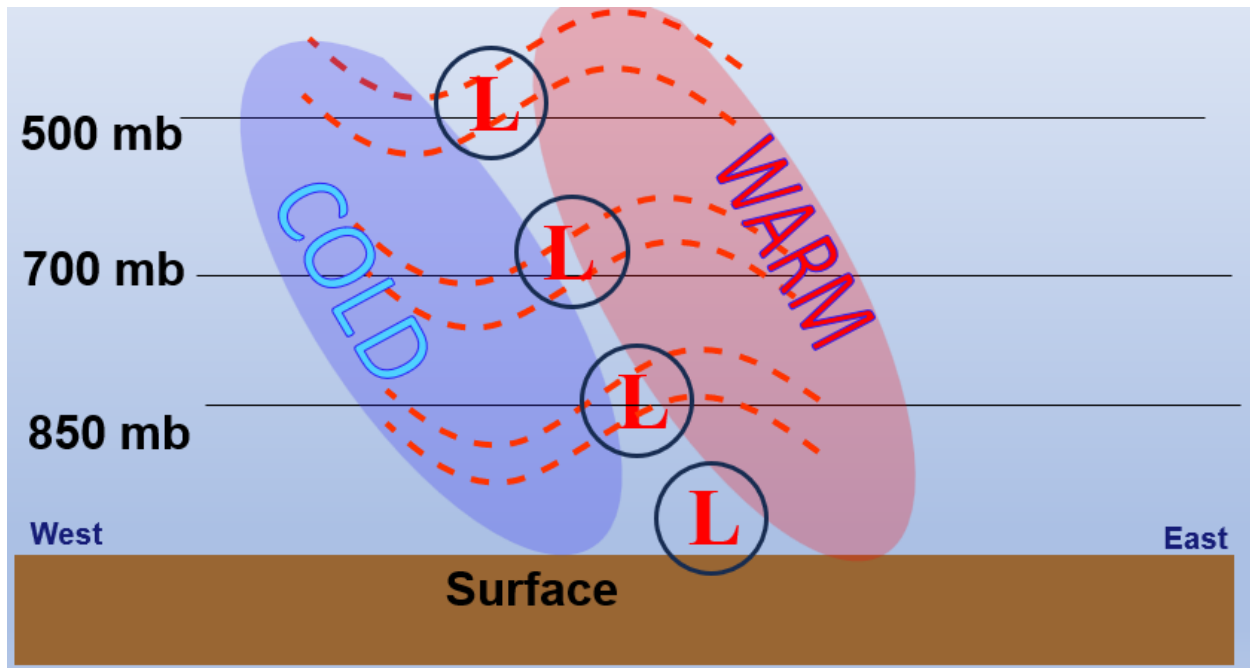
Barotropic vs. Baroclinic

Barotropic: In barotropic weather systems, temperature patterns (isotherms) closely match pressure patterns (isobars), and height patterns (contours), indicating minimal temperature advection. These systems are shallow, lacking influence from upper-level winds. Because they are vertically aligned and not steered by upper-level features, barotropic systems tend to move very slowly.



Baroclinic: In contrast to barotropic systems, baroclinic weather systems exhibit temperature patterns (isotherms) that are more perpendicular to pressure patterns (isobars) and height patterns (contours), indicating significant temperature advection. Surface features in these systems are strongly linked to and supported by upper-level atmospheric features. Temperature changes

occur throughout the atmosphere, resulting in faster movement of the system in a downstream direction.



Exercise Questions: Advection / Barotropic / Baroclinic

Convergence, Divergence, and Vertical Motions

Vertical motions refer to the upward or downward movement of air parcels in the atmosphere, driven by factors like temperature differences, pressure gradients, and convergence/divergence patterns.

Convergence: Convergence occurs when winds collide, resulting in an accumulation of air mass. This excess air is then forced to move vertically. Specifically, convergence at the surface primarily results in upward vertical motion as warm air is forced aloft. Conversely, convergence aloft typically forces cold air downwards, leading to downward vertical motion.

Divergence: Divergence happens when winds spread apart, creating a "void" or removal of air mass. To fill this void, air from above and below must sink and rise, respectively. This means warmer air from below will rise (upward vertical motion) and colder air from above will sink (downward vertical motion) to compensate for the spreading air.

Chimney Effect: The "Chimney Effect" refers to the upward vertical motion of warm air caused by either lower-level convergence, or upper-level divergence. The atmospheric effects of the chimney effect results in surface pressure decreasing and upper-level heights to rise as the warm air expands the atmosphere above it.

Damper Effect: The "Damper Effect" refers to the downward vertical motion of cold air caused by either lower-level divergence, or upper-level convergence. The atmospheric effects of the damper effect results in surface pressure rising and upper-level heights to fall as the cold air compresses the atmosphere above it.

Level Of Non-Divergence (LND): Whether you analyze convergence or divergence at a point in the atmosphere, the converging or diverging air parcels must switch to the opposite at some point to maintain the chimney or damper effect. The Level of Non-Divergence refers to the area of the atmosphere where the convergence or divergence switches to the opposing effect. The LND is typically found at the halfway point in either chimney or damper effect as the parcels are either rising or sinking. EX. If convergence occurs at the surface and extends up to 500mb ($\approx 18000\text{ft}$), then $\approx 9000\text{ft}$ is where the convergence would start becoming divergence aloft.

Exercise Questions: Convergence / Divergence / Vertical Motion

Review Questions: Comprehensive Physics & Dynamics Review

40Q Appraisal: Physics & Dynamics

Introduction to Macroscale Analysis

The goal of this objective will be to show you have the capability to perform a basic macroscale analysis. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly identify and label various weather features discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

Global Geography

Landforms and Bodies of Water

The Earth is divided into seven continents: Africa, Antarctica, Asia, Europe, North America, South America and Oceania. and five oceans: Atlantic, Pacific, Indian, Arctic, and Southern. Familiarize yourself with different landforms like mountains, plains, and deserts, and bodies of water like rivers, lakes, and oceans. Understanding geographic information at different scales, from local to global, is important.

Latitude and Longitude

Latitude and longitude are a grid system used to pinpoint locations on Earth.

Latitude

Latitude measures how far north or south a point is from the equator. Latitude is depicted using a system of parallel lines that run east to west. Latitude is measured in degrees ranging from 0° at the equator to 90°N at the North Pole and 90°S at the South Pole. The equator is the line that separates the northern and southern hemispheres.

Longitude

Longitude measures how far east or west a point is from the Prime Meridian. Longitude is depicted using a system of parallel lines that run north to south. Longitude is also measured in degrees ranging from the Prime Meridian (0°) which runs through Greenwich, England to the [International Date Line](#) (180°) which is the line that separates the eastern and western hemispheres.

For example, the coordinates for Keesler AFB would be 30.4097° N, 88.9262° W. This tells us we are 30° North of the Equator, and 88° West of the Prime Meridian.

Every location on earth has a unique latitude and longitude. Understanding how to read these coordinates is a crucial skill for weather forecasters to have. Without being able to properly locate or communicate specific locations, you could be analyzing or forecasting for the wrong area which could lead to catastrophic results for those executing the mission.

Exercise / Practice: Geography / Latitude / Longitude

Identifying Macroscale Weather Features

Macroscale Definition

Weather features that are best viewed on a continental to global or hemispheric scale.

ADD SOP Unit 3 Obj 3a. – Weather Chart Analysis SOP

Air Masses

Macroscale air masses are expansive bodies of air that form over large, uniform source regions and influence weather across entire continents and ocean basins. The two macroscale air masses are **Continental Arctic (cA)** and **Maritime Tropical (mT)**.

Each air mass is identified using a **two-letter code**:

- The **first letter** describes the moisture source:
 - **c** = Continental → dry air formed over land
 - **m** = Maritime → moist air formed over water
- The **second letter** describes the thermal characteristic:
 - **A** = Arctic → frigid polar origin
 - **T** = Tropical → warm equatorial origin

Continental Arctic (cA)

The Continental Arctic (cA) air mass forms over extremely cold land surfaces typically located poleward of 60° latitude in either hemisphere. These source regions include areas such as Siberia, Greenland, and parts of Antarctica. This air mass reaches its greatest intensity and spatial coverage during the winter season. the summer months, cA air weakens and retreats toward the poles.

cA air masses are responsible for polar outbreaks, cold-core high pressure systems, and extreme winter conditions. Their identification is crucial for understanding cold air advection and the structure of polar air intrusions in both hemispheres.

Key Characteristics:

- **Temperature:** Extremely Cold
- **Moisture:** Very dry
- **Stability:** Extremely Stable

Chart Indicators:

- **Thickness:** Typically, ≤ 528 dam on the thickness chart.
- **Geography:** Located over land masses.

Maritime Tropical (mT)

The Maritime Tropical (mT) air mass forms over warm tropical and subtropical oceans located between the equator and approximately 35° latitude in both hemispheres. Source regions include the Caribbean Sea, western Pacific, Indian Ocean, and tropical South Atlantic. These regions provide abundant heat and moisture, allowing the air mass to become warm, humid, and unstable. mT air expands poleward during the summer season in each hemisphere, bringing hot, moist conditions and supporting convective activity. In winter, it retreats equatorward and weakens.

mT air masses are the primary source of moisture and instability for warm-season weather systems across both hemispheres. They fuel thunderstorms, tropical cyclones, and frontal boundaries, making them essential to identify when analyzing moisture transport and convective potential.

Key Characteristics:

- **Temperature:** Hot Temperatures
- **Moisture:** Moist / Fully Saturated
- **Stability:** Extremely Unstable

Chart Indicators:

- **Thickness:** Typically, ≥ 570 dam on the thickness chart.
- **Geography:** Located over large bodies of water.

Three Cell Model

Meteorologists use a three-cell model to describe how and where air moves within our global circulation. The three cells are from the poles to equator in both hemispheres are the Polar, Ferrel, and Hadley cells.

Polar Cell: The Polar Cell begins with a damper effect near the poles with convergence aloft and divergence at the surface and extends equatorward with a chimney effect near 60° latitude with convergence at the surface and divergence aloft. The Polar Cell has the lowest tropopause out of the three cells due to the cold polar air compressing the atmosphere to much lower heights than the Ferrel or Hadley Cell usually near 300mb. The Polar Cell contains the continental arctic airmass.

Ferrel Cell: The Ferrel cell begins with the chimney effect near 60° latitude with convergence at the surface and divergence aloft and extends equatorward with a damper effect near 30° latitude with divergence at the surface and convergence aloft. The Ferrel Cell has a tropopause height somewhere between the Polar and Hadley Cell usually near 200mb.

Hadley Cell: The Hadley cell begins with a damper effect near 30° latitude with divergence at the surface and convergence aloft and extends equatorward with a chimney effect near the equator with convergence at the surface and divergence aloft. The Hadley Cell has the highest tropopause out of the three cells due to the warm tropical air near the Equator expanding the atmosphere to much higher heights than the Polar or Ferrel Cell usually near 100mb. The Hadley Cell contains the maritime tropical airmass.

Semi-Permanent Pressure Systems

With the interaction of the three cells that form our atmosphere, the resulting upward and downward vertical motions create different barotropic pressure centers at the surface we refer to as Semi-Permanent Pressure Systems. These systems fall into one of four categories.

Polar High: The damper effect near the poles creates a Continental Arctic (cA) air mass and high-pressure center called the Polar High. This is a cold core system meaning cold air extends throughout the atmosphere from the surface to the tropopause resulting in a low height center above the surface high pressure center.

Subpolar Lows: The chimney effect near 60° latitude creates low pressure centers at the surface called subpolar lows. There are two in the northern hemisphere we call the Aleutian low and the Icelandic low. Subpolar lows are not air masses but are still considered cold core system which results in a low height center above the surface low pressure center. Subpolar Lows will migrate with the seasons. They will move West in the Winter and East in the Summer.

Subtropical Highs: The damper effect near 30° latitude creates a high-pressure center at the surface called subtropical highs. The Northern hemisphere has two subtropical highs called the

North Pacific high and the Bermuda-Azores high. The Subtropical highs are maritime tropical (mT) air masses, as well as warm-core systems resulting in a high height center above the surface low pressure center. Subtropical Highs also migrate with the seasons. They move opposite the Subpolar Lows. Subtropical Highs will move West in the Summer, and East in the Winter.

Equatorial Lows: The chimney effect near the Equator creates equatorial low-pressure centers at the surface. These are warm core systems that are more prevalent in the northern hemisphere during the summer and the southern hemisphere during winter.

Exercise / Practice: Air Masses / 3-Cell Model / Semi-Perms

Jet Streams and the Long-Wave Pattern

Jet streams are strong horizontal winds concentrated in narrow bands located in the upper atmosphere. They are thousands of miles long, hundreds of miles wide, and two to three miles thick. There are two terms associated with jet streams:

Jet Stream: A band of wind speeds greater than or equal to 50 knots analyzed by isotachs.

Jet Axis or Jet Core: An interconnecting band of the highest wind speeds near the center of a jet stream and is depicted on a chart by a solid red line.

Formation Requirements: Jet Streams are created by the interaction of the three cells as well as strong horizontal temperature contrast. For example, there is relatively colder air on the poleward side and relatively warmer air on the equatorward side of each jet stream. The temperature contrast is measured perpendicular to the jet axis. If the temperature contrasts are relatively weak, it will be a weaker jet stream. If the temperature contrast is greater, a stronger jet stream would form.

Three Tropospheric Jets

Arctic jet. The Arctic jet is associated with the Polar cell. It forms near the Arctic Circle and has the lowest altitude of all three jets typically found near 400mb. Due to a weaker temperature contrast in the cA air, the AJ is weaker than the PFJ.

Polar front jet (PFJ). The PFJ is located at the intersection of the Polar and Ferrel cells. It varies in altitude and temperature more than the other two jets due to the largest temperature change between seasons. The PFJ is found using the 300mb chart and is in large bands of wind speeds of 50KT or more.

Subtropical jet (STJ). The STJ is located at the intersection of the Ferrel and Hadley cells. The STJ has the highest altitude of all three jets, typically found near 150mb. Due to a weaker temperature contrast in the mT air, the STJ is weaker than the PFJ.

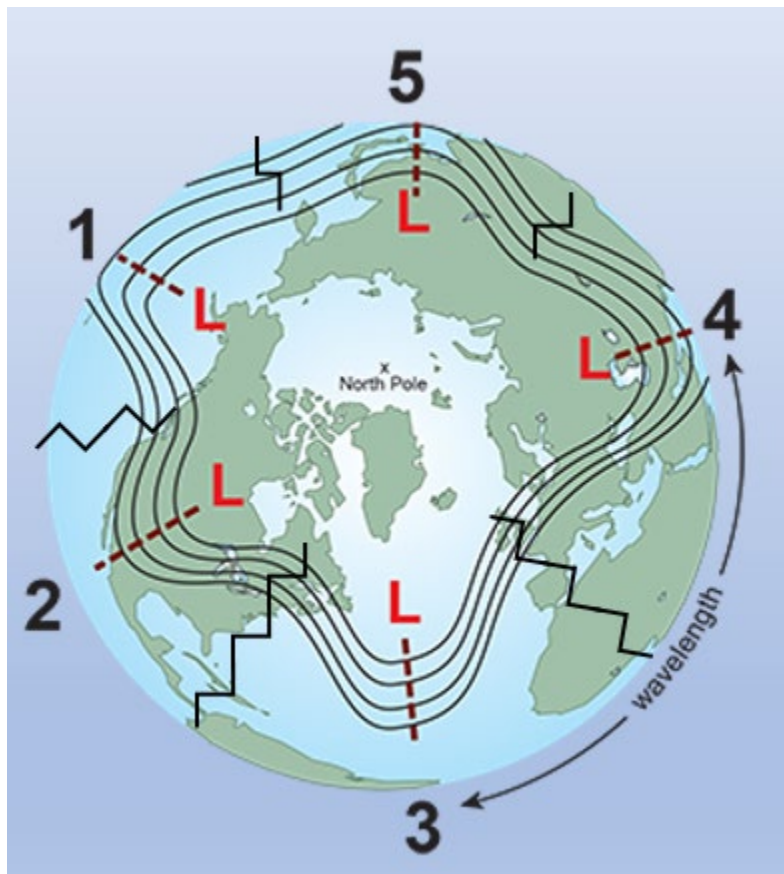
Long Wave Pattern

The Long Wave Pattern is an upper air feature that is typically found between 25°N and 65°N Latitude and describes the overall upper-level wind flow that is usually associated with the Polar Front Jet. The Long Wave pattern is analyzed on a 500mb chart and can be found by finding the strongest contour gradient between the 552 and 570 decameter contour lines.

Formation principals The Long Wave Pattern is made up of **macroscale ridges and troughs** that flow **west to east** around the globe in the northern hemisphere:

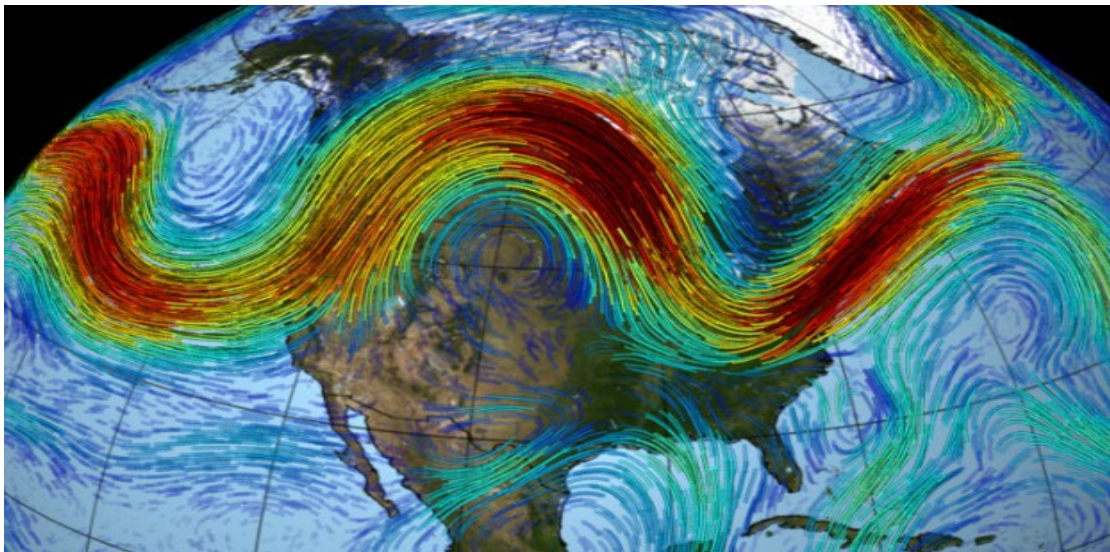
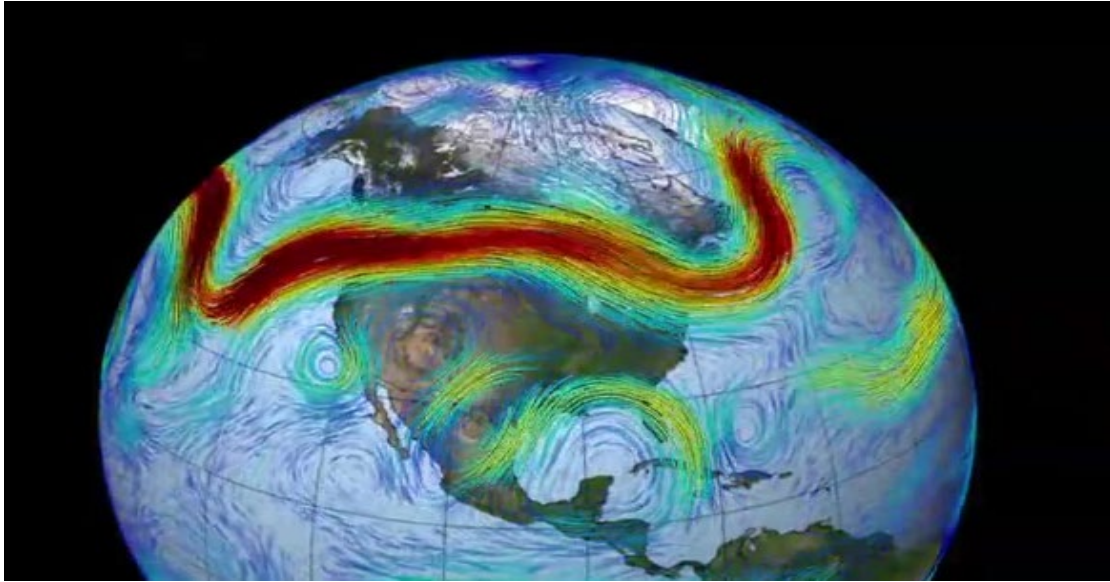
- **Long Wave Ridges:** Poleward extensions of **warm, maritime tropical (mT)** air from the equator and subtropical highs
- **Long Wave Troughs:** Equatorward extensions of **cold, continental arctic (cA)** air from the poles and polar highs

The transition point between a trough and a ridge — where cyclonic flow begins turning into anticyclonic flow — is called the **Inflection Point**.



Zonal vs. Meridional: There are two types of long-wave patterns: zonal and meridional. In a zonal pattern, airflow is primarily flowing west to east, whereas a meridional pattern will have more north to south flow. Regardless of the pattern, it separates cold, arctic air from warm, tropical air in the upper atmosphere. On a planetary scale, seasonal changes and the interaction of smaller scale weather features moving along the long-wave pattern are the reasons the long-wave pattern will change between zonal and meridional flow. During summer in the northern

hemisphere, the long-wave pattern is primarily zonal due to warmer temperatures being able to extend more poleward, increasing atmospheric heights. The pattern changes to meridional in the other seasons due to a decrease in temperature allowing the cold, polar air to extend more equatorward, decreasing atmospheric heights.



Exercise / Practice: Long-Wave Pattern

Practice: Macro Analysis

PC: Macro Analysis